

What is the probability that a Poisson random variable with mean $\ln 2$ is less than $3/2$?

$$p_X(X=k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k=0, 1, 2, \dots, n$$

(a) $\ln \sqrt{2e}$

(b) $1/\ln(2e)$

(c) $1/\ln(e^2)$

(d) $\ln \sqrt{2}$

(e) $\ln \sqrt{e}$

(f) $1/e$

(g) e^{-2}

(h) $2e^{-2}$

(i) $1/(e \ln 2)$

(j) $1/(2 \ln 2)$

(k) $1/2$

(l) $1/4$

(m) None of these

$$E[X] = \lambda = \ln(2)$$

$$P(X < \frac{3}{2}) = P(X=0) + P(X=1)$$

$$= \frac{(\ln(2))^0 e^{-\ln(2)}}{0!} + \frac{(\ln(2))^1 e^{-\ln(2)}}{1!}$$

$$= e^{\ln(2^{-1})} + \ln(2) \cdot \frac{1}{2}$$

$$= \frac{1}{2} + \ln(2) \cdot \frac{1}{2}$$

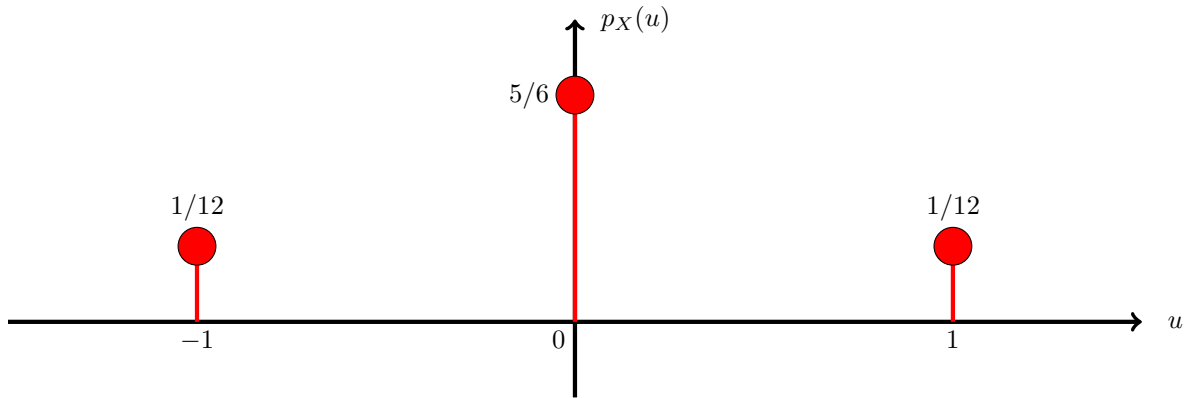
$$= \frac{1}{2} (\ln(2) + 1) \leftarrow \ln(e) = 1$$

$$= \frac{1}{2} (\ln(2) + \ln(e)) \quad \ln(ab) = \ln(a) + \ln(b)$$

$$= \frac{1}{2} (\ln(2e)) \quad a \ln(b) = \ln(b^a)$$

$$= \ln(2e^{\frac{1}{2}})$$

If X is a discrete random variable whose probability mass function is shown below, then what is the variance of $2 - 3X^2$?



$$\text{Var}(2 - 3X^2)$$

$$\text{let } Y = 2 - 3X^2$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2$$

$$\text{@ } x = -1, Y = -1 \quad P(Y = -1) = \frac{1}{12} = \frac{1}{6}$$

$$\text{@ } x = 0, Y = 2 \quad P(Y = 2) = \frac{5}{6}$$

$$\text{@ } x = 1, Y = -1$$

$$E[Y] = -1\left(\frac{1}{6}\right) + 2\left(\frac{5}{6}\right)$$

$$= \frac{9}{6} = \frac{3}{2}$$

$$E[Y^2] = (-1)^2\left(\frac{1}{6}\right) + (2)^2\left(\frac{5}{6}\right)$$

$$= \frac{1}{6} + \frac{20}{6}$$

$$= \frac{21}{6} = \frac{7}{2}$$

$$\text{Var}(Y) = \frac{7}{2} - \left(\frac{3}{2}\right)^2 = \frac{7}{2} - \frac{9}{4} = \frac{5}{4}$$

(a) 5/4

(b) 5/8

(c) 3/2

(d) 3/4

(e) 1/2

(f) 4/5

(g) 3/5

(h) 2/25

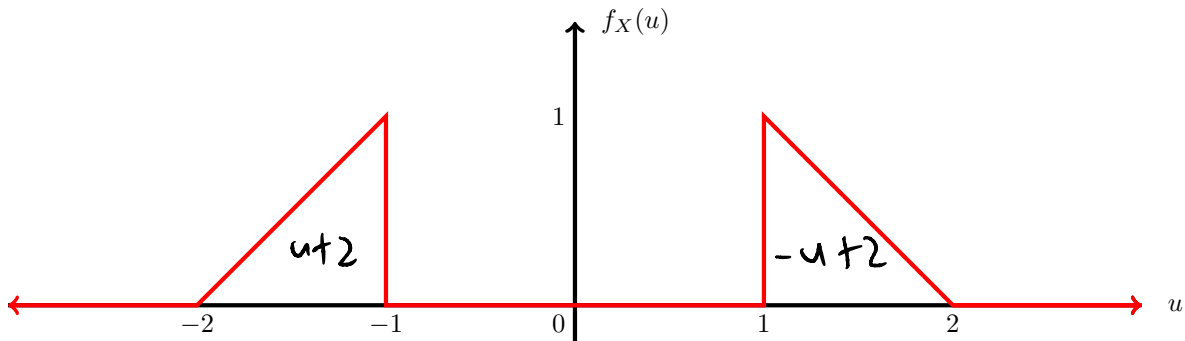
(i) 1/25

(j) 9/25

(k) 3/25

(l) None of these

What is the variance of a random variable X whose probability density function is shown below?



$$E[X] = 0$$

$$E[X^2] = \int_{-\infty}^{\infty} u^2 f_X(u) du$$

(a) 11/6

(b) 11/12

(c) 3/4

(d) 5/6

(e) 7/12

(f) 1/6

(g) 1/3

(h) 0

(i) 1/2

(j) 1/8

(k) 5/2

(l) None of these

$$2 \cdot \text{right triangle} = 2 \int_1^2 u^2 (-u+2) du$$

$$= 2 \int_1^2 (-u^3 + 2u^2) du$$

$$= -2 \int_1^2 u^3 du + 4 \int_1^2 u^2 du$$

$$= -2 \left[\frac{1}{4} u^4 \right]_1^2 + 4 \left[\frac{1}{3} u^3 \right]_1^2$$

$$= -\frac{1}{2} [16 - 1] + 4 \left[\frac{8}{3} - \frac{1}{3} \right]$$

$$= -\frac{15}{2} + \frac{28}{3}$$

$$= -\frac{45}{6} + \frac{56}{6} = \frac{11}{6}$$